Day 19 | Tuesday

19

December 1, 2026

Volume of boxes and prisms

12:45 PM–2:00 PM • Math • 75 min

Cylinders, capacity and missing heights

2:15 PM–3:30 PM • Math • 75 min

Week 6 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Volume of boxes and prisms

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 6, Lesson 3

2:15 PM–3:30 PM • 75 min

Cylinders, capacity and missing heights

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 6, Lesson 4

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Volume of boxes and prisms

Today’s goal: Calculate prism volume using consistent units and convert a capacity to litres.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Exercise Book 11A Q1–10 (p.103); 11B Q1,5 (p.105)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Recall and reconnect

Try to distinguish a length, surface area and volume; identify the third dimension on a box.

Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 11A Q1–10 (p.103); 11B Q1,5 (p.105)

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

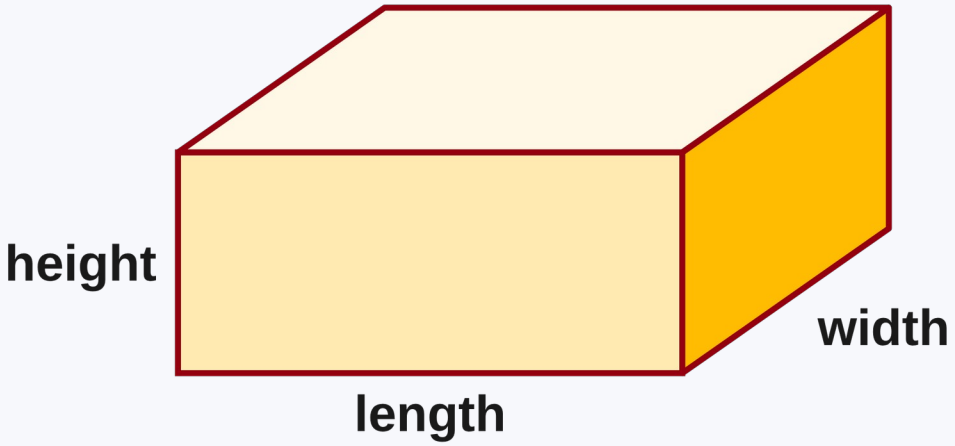
Volume uses three dimensions

1 For a rectangular prism, V = length × width × height.

2 Convert all three lengths to the same unit first.

3 The result uses cubic units because three lengths were multiplied.

4 Doubling only one dimension doubles the volume.



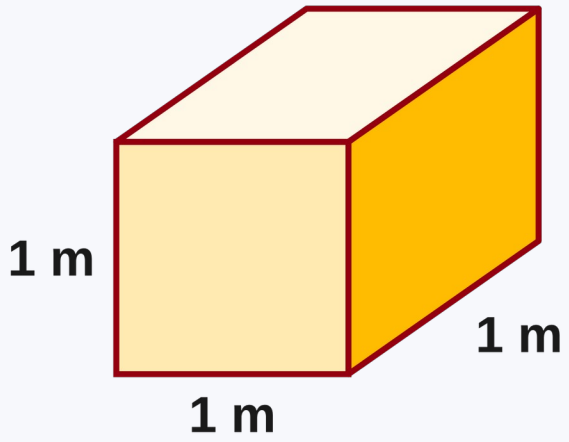
V = length × width × height

three lengths: cubic units

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Connect volume and capacity

1. Capacity describes how much a container can hold.
2. 1 m³ = 1,000 L; use the book’s matching capacity conversion.
3. A cubic length conversion applies the length factor three times.
4. For pails, divide by one-pail capacity and round as the purchase requires.



1 m³ = 1,000 L

capacity: how much it holds

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Keep dimensions consistent before multiplying

Reservoir example • 800 mm × 500 mm × 450 mm
Convert to metres: 0.8 m × 0.5 m × 0.45 m.

$V = 0.8 \times 0.5 \times 0.45 = 0.18\ m^3$
Capacity = 0.18 × 1,000 = 180 L
20 L pails needed = 180 ÷ 20 = 9

Check • 9 × 20 L = 180 L. The unit tells us what the number measures.
Doubling one dimension doubles the volume; doubling all three multiplies it by 8.

Together • Work 11A Q2–7,9. Estimate magnitude before calculating.
Shape Canvas: show all three prism dimensions and test a doubled length.

Exercise Book 11B Q1 p.105 • Numeracy Hub → Volume and Layout → Volume and capacity

Textbook | page 269

VOLUME OF A PRISM

V = end area × length

Box: V = l × w × h

Find the end area with the formulas from Chapter 10. All lengths in the same unit.

Textbook p.269 • Volume of a prism — End area times length

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Textbook | page 269

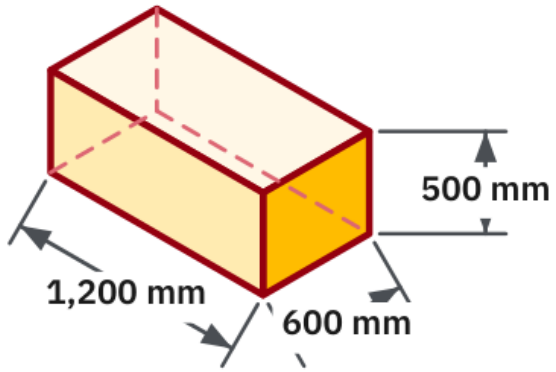
Example 11.1 A rectangular tank

A tank is 1,200 mm long, 600 mm wide and 500 mm high inside.

Find its volume in mm³ and in m³.

SOLUTION

1. $V = l \times w \times h = 1{,}200 \times 600 \times 500 = 360{,}000{,}000$ mm³.
2. In metres the tank is 1.2 m by 0.6 m by 0.5 m.
3. $V = 1.2 \times 0.6 \times 0.5 = 0.36$ m³.



Textbook p.269 • Example 11.1 — Three dimensions of a rectangular tank

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Spot the mistake. Explain the correction.

A box uses metres for two sides and millimetres for the third. Can the numbers be multiplied as shown?

Discuss the method before checking the answer.

Convert every dimension to one length unit first, then multiply and label the cubic result.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Work together

Work Exercise Book 11A Q2–7 and Q9. Learners estimate magnitude before calculating.

Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 11A Q1–10 (p.103); 11B Q1,5 (p.105)

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Guided practice | 11A Q2

Write 3 cm^3 in mm^3.

Work with a partner. Take turns explaining the next step.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Compare your method | 11A Q2

Write 3 cm³ in mm³.

CHECK TOGETHER

3,000 mm³

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Your scheduled practice

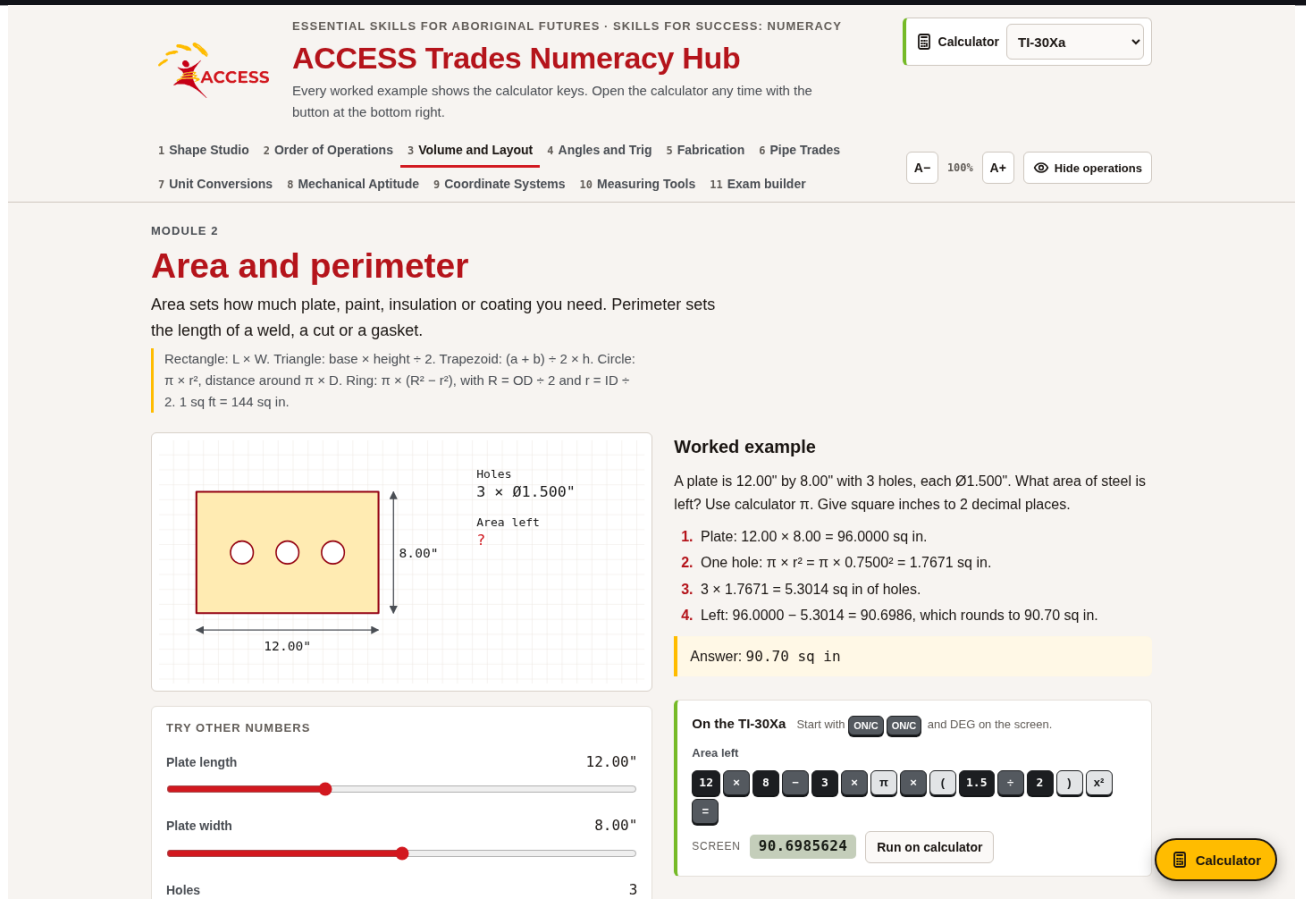
Complete Exercise Book 11A Q8,10 and 11B Q1,5. Use a Shape Canvas prism to show how doubling one dimension changes volume.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 11A Q1–10 (p.103); 11B Q1,5 (p.105)

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Practise in Numeracy Hub and Shape Canvas

1. Create a rectangular prism in Shape Canvas and display all three dimensions.

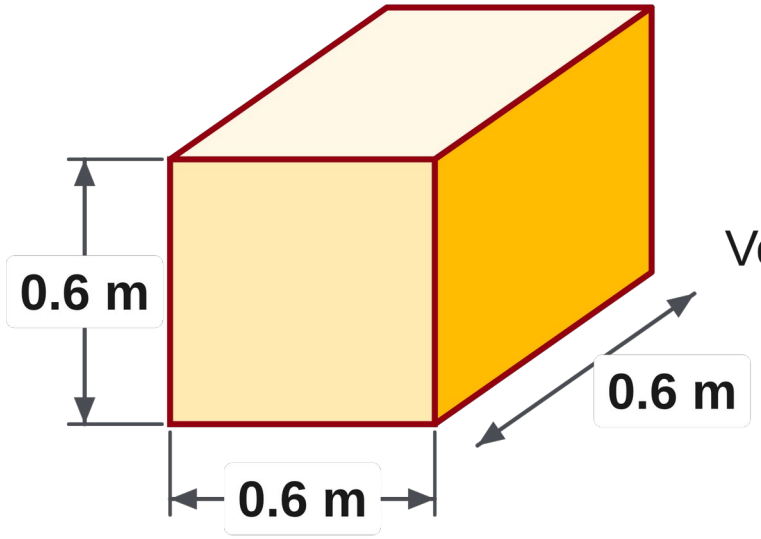
2. Predict and calculate the volume change when only one dimension is doubled.

3. Check a box/slab example in Numeracy Hub Volume and Layout → Volume and capacity.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Practise from the Exercise Book | 11A Q8

Cube with 0.6 m edges. Find the volume in m³.



Volume (m³) ?

Write the operation, working and units before checking.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Practise from the Exercise Book | 11A Q10

Write 1,155 in³ in US gallons.

Write the operation, working and units before checking.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Check independently

Exercise Book 11A Q7 and Exercise Book 11B Q1; label volume/capacity and whole pails.

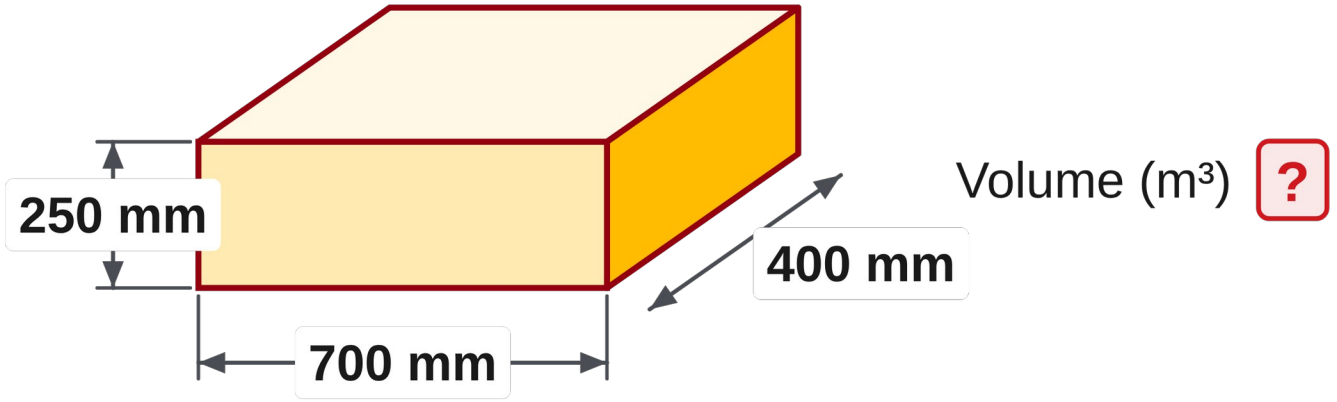
Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 11A Q1–10 (p.103); 11B Q1,5 (p.105)

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Independent check | 11A Q7

Box 700 mm by 400 mm by 250 mm. Find the volume in m³.



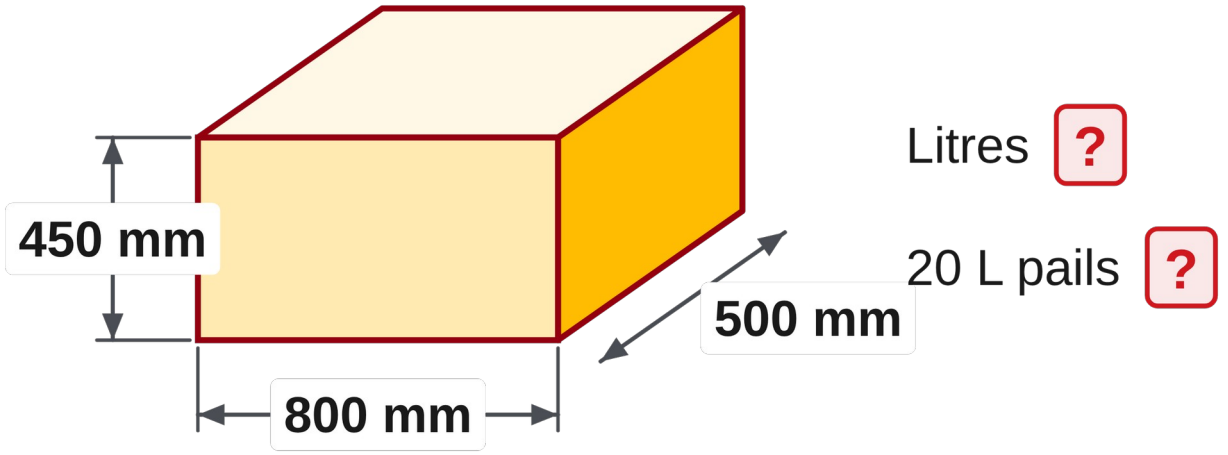
Volume (m³) ?

Answer independently. Keep worked solutions closed.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Independent check | 11B Q1

A hydraulic reservoir is 800 mm by 500 mm by 450 mm inside. How many litres does it hold? How many 20 L pails of oil fill it?



Litres ?

20 L pails ?

Answer independently. Keep worked solutions closed.

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 11A Q1–10 (p.103); 11B Q1,5 (p.105)

Textbook sections 11.1–11.3 pp.268–272; sections 7.4 pp.170–172

Break

2:00 PM–2:15 PM

Return ready for: Cylinders, capacity and missing heights

This break is excluded from teaching minutes.

Cylinders, capacity and missing heights

Today’s goal: Find a cylinder’s volume and use a known capacity to solve for a missing height or length.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Exercise Book 11A Q11–13,19,21–22 (p.104); 11B Q2 (p.105); 11C Q1–2 (p.107)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Recall and reconnect

Explain why diameter must be halved before using $\pi r^2 h$; distinguish US and imperial gallons.

Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 11A Q11–13,19,21–22 (p.104); 11B Q2 (p.105); 11C Q1–2 (p.107)

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

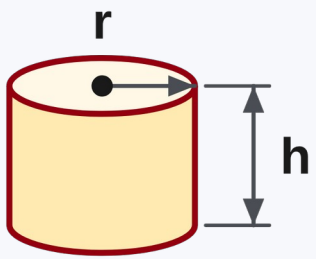
Find the volume of a cylinder

1 A cylinder has circular base area πr^2.

2 Multiply base area by height: $V = \pi r^2 h$.

3 If the drawing gives diameter, halve it before using the radius formula.

4 Use matching length units, then convert the resulting volume to capacity.



$V = \pi r^2 h$

base area πr^2 × height

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Recover a missing height from volume

1. Write $h = V \div (\pi r^2)$ for a cylinder.
2. Keep the full base-area expression inside the denominator brackets.
3. Use a consistent volume unit before dividing by the base area.
4. Substitute the found height into the original volume formula to check.

$h = V \div (\pi r^2)$
write it first → **Brackets kept**
round πr^2 → **Same volume**
unit → **Check**
$\pi r^2 h = V$

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Divide volume by base area to find height

Cylinder example • r = 150 mm; h = 600 mm.
$V = \pi \times 150^2 \times 600 \approx 42{,}411{,}501\ mm^3 \approx 42.41\ L$.
Check the factor: 1 L = 1,000,000 mm^3.

Missing-height example • Rectangular base: 600 mm × 500 mm.
240 L = 240,000,000 mm^3.
h = 240,000,000 ÷ (600 × 500) = 800 mm.

Together • Work 11A Q13,19 and 11C Q1.
Check each height by multiplying base area × height.
Shape Canvas: hide a box or cylinder height; solve before revealing it.

Exercise Book pp.104,107 • Numeracy Hub Volume and capacity; Unit Conversions for explicitly labelled L/US gal

Textbook | page 273

11.4 Cylinders: tanks and pipes

A **cylinder** has two equal circular ends joined by a curved side. It works like a prism: every slice is the same circle. So its volume is the circle area times the height (or length).

VOLUME OF A CYLINDER

$$V = \pi \times r^2 \times h$$

r = radius of the end, h = height or length. For a pipe, h is the length of the run.

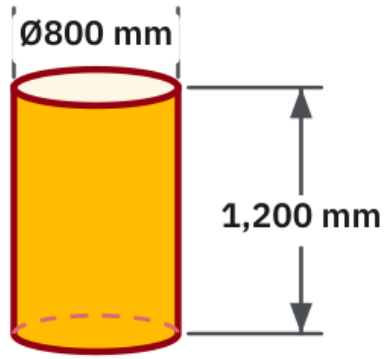
Textbook p.273 • Cylinder volume — Circular base area times height

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Textbook | page 274

Example 11.6 A vertical tank

A vertical tank is 800 mm across inside and 1,200 mm high. How many litres does it hold, to two decimal places?



SOLUTION

1. Radius: 800 ÷ 2 = 400 mm.
2. $V = \pi \times 400^2 \times 1{,}200 = 603{,}185{,}789.49\ mm^3$.
3. Litres: ÷ 1,000,000 gives 603.19 L.

Textbook p.274 • Example 11.6 — Cylinder dimensions and capacity

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Spot the mistake. Explain the correction.

Does a gallon conversion work the same for US and imperial gallons?

Discuss the method before checking the answer.

No. Identify which gallon the question specifies and select its matching conversion factor.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Work together

Work Exercise Book 11A Q13,19 and Exercise Book 11C Q1. Partners check by resubstitution.

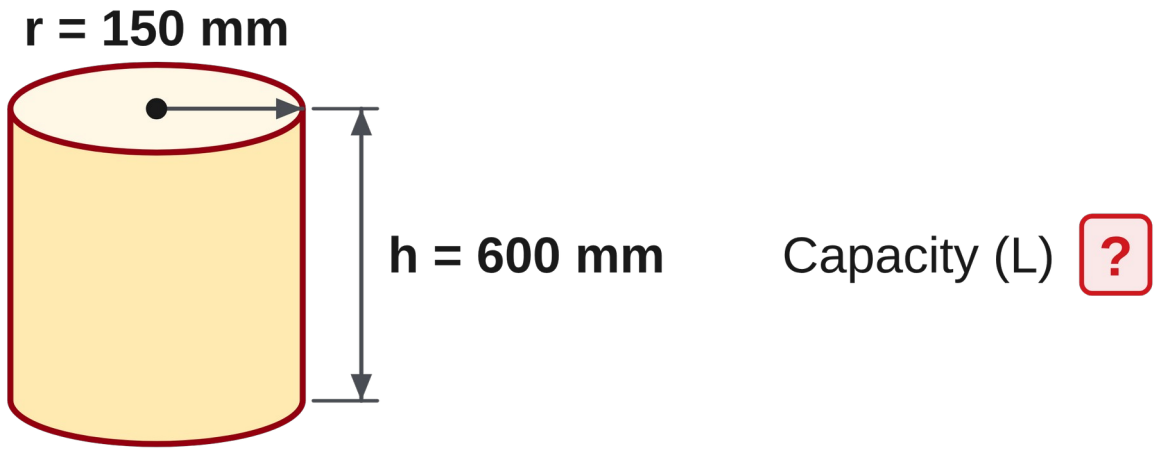
Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 11A Q11–13,19,21–22 (p.104); 11B Q2 (p.105); 11C Q1–2 (p.107)

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Guided practice | 11A Q13

Cylinder: radius 150 mm, height 600 mm. Capacity in litres, to two decimal places?



Work with a partner. Take turns explaining the next step.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Compare your method | 11A Q13

Cylinder: radius 150 mm, height 600 mm. Capacity in litres, to two decimal places?

CHECK TOGETHER

42.41 L

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Your scheduled practice

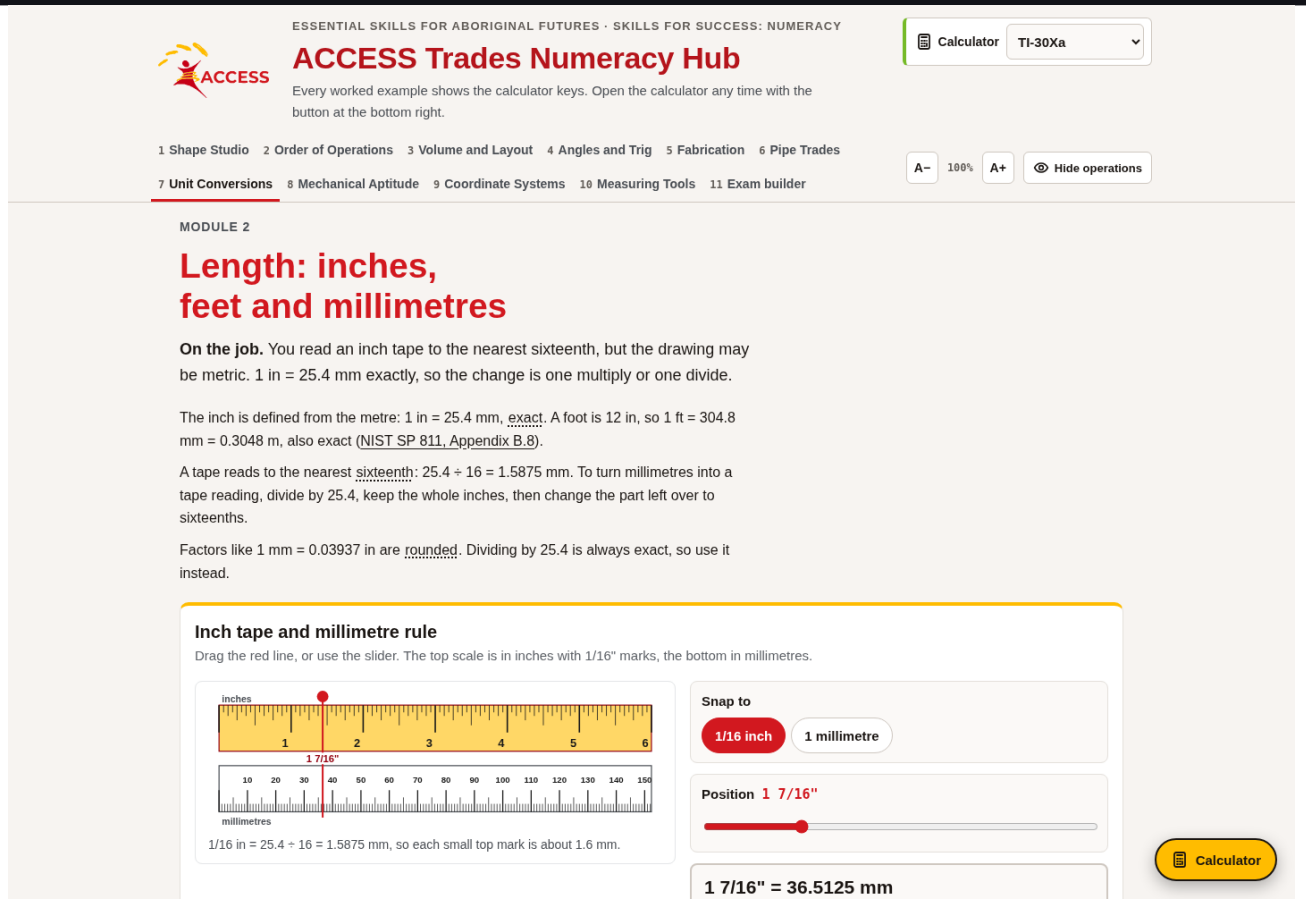
Complete Exercise Book 11B Q2 and 11C Q2; compare 11A Q21. Use Shape Canvas to verify a hidden dimension. Cones/spheres are optional book extension, not required this session.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 11A Q11–13,19,21–22 (p.104); 11B Q2 (p.105); 11C Q1–2 (p.107)

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Practise in Numeracy Hub and Shape Canvas

1. Set up a cylinder or box in Shape Canvas, then hide its height.

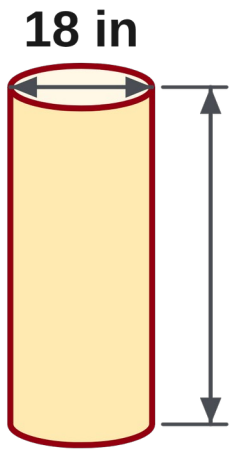
2. Calculate the height on paper from the stated volume or capacity and consistent units.

3. Check in Volume and capacity; use Unit Conversions for the specified L/US gal conversion.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Practise from the Exercise Book | 11B Q2

A water heater tank is a cylinder 18 in across and 48 in tall inside. Find its capacity in US gallons, to two decimal places.



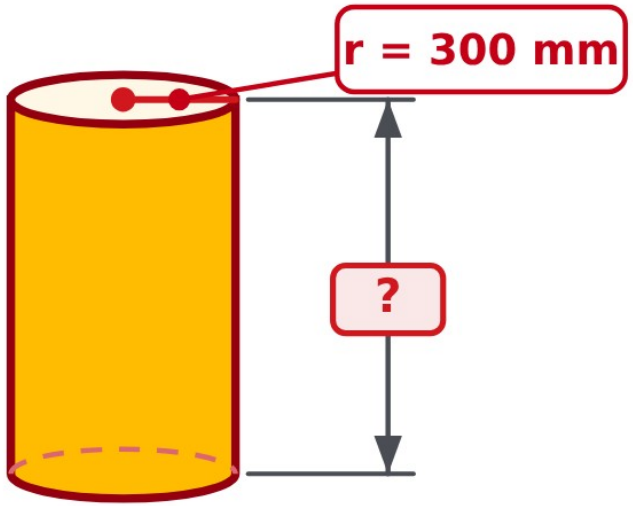
Capacity (US gal) ?

Write the operation, working and units before checking.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Practise from the Exercise Book | 11C Q2

The cylinder holds 282.74 L to two decimal places. Find the hidden height to the nearest mm.



Write the operation, working and units before checking.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Check independently

Exercise Book 11A Q13 and Exercise Book 11C Q1; include capacity units.

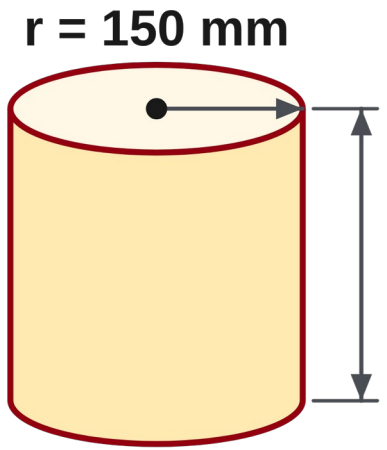
Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 11A Q11–13,19,21–22 (p.104); 11B Q2 (p.105); 11C Q1–2 (p.107)

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Independent check | 11A Q13

Cylinder: radius 150 mm, height 600 mm. Capacity in litres, to two decimal places?



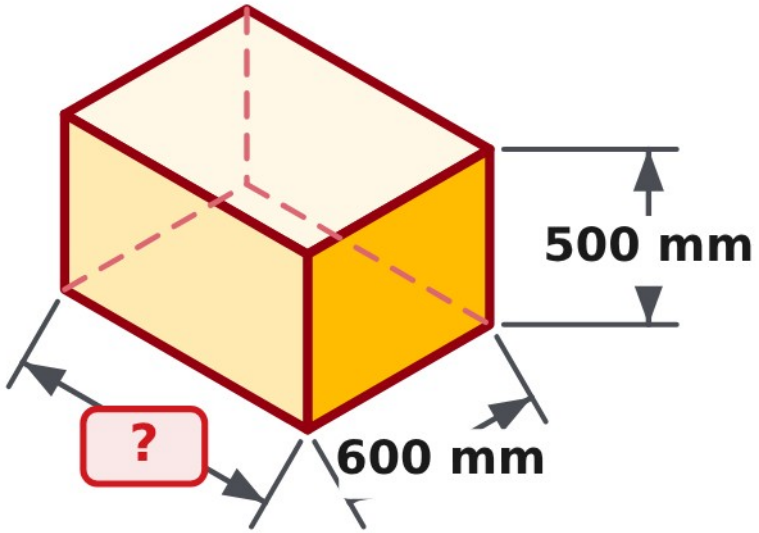
Capacity (L) ?

Answer independently. Keep worked solutions closed.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Independent check | 11C Q1

The tank holds 240 L. Find the hidden length.



Answer independently. Keep worked solutions closed.

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 11A Q11–13,19,21–22 (p.104); 11B Q2 (p.105); 11C Q1–2 (p.107)

Textbook sections 11.3–11.4 pp.271–274; sections 8.6 pp.201–203